LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flow over bodies; bluff bodies and streamlined bodies, Flow pattern over flat plate, formation of eddies, the wake and its reduction.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flow over bodies; bluff bodies and streamlined bodies, Flow pattern over flat plate, formation of eddies, the wake and its reduction.
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flow visualization diagrams; Safety: No physical handling of models; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of a fluid?
(b) Define viscosity in the context of air flow.
(c) What is the primary difference between a liquid and a gas regarding compressibility?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Streamlined vs Bluff bodies • Flow pattern over flat plate • Boundary layer formation	L & Ds	PPT, IB	15 min
(b)	• Formation of eddies • Wake characteristics • Drag reduction techniques	L & Ds	PPT, PDF	20 min
(c)	• Comparison of flow types • Impact on aircraft performance • Summary of key concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a bluff body and a streamlined body.
(b) Describe how eddies form in the wake of a flat plate.
(c) How does boundary layer separation affect drag?

12. Home Assignment and Follow Up	Prepare a brief note on how understanding flow over bodies aids in the design of efficient aircraft wings, linking to concepts covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Boundary layer, laminar, transition, turbulent B/L,	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min